Software Project Management
Assignment-7
TYPE OF QUESTION: MCQ/MSQ

Number of questions: **10** **Total mark: 10 X 1 = 10**

QUESTION 1:

If the optimistic estimate is 4 days, the most likely estimate is 10 days, and the pessimistic estimate is 16 days, which of the following statements are correct?

a. Expected time is 10 days.
b. Variance is 4.
c. Standard deviation is 2.
d. Expected time is 12 days.
e. Variance is 2.

Correct Answer: a. Expected time is 10 days.
b. Variance is 4.
c. Standard deviation is 2

Detailed Solution:

Expected time (TE) = (O + 4M + P) / 6 = (4 + 4×10 + 16) / 6 = 60/6 = 10 days → (a) is true, (d) is false.
Variance (V) = ((P – O) / 6)² = ((16 – 4) / 6)² = (12/6)² = 2² = 4 → (b) is true, (e) is false.
Standard deviation (SD) = √V = √4 = 2 → (c) is true.

QUESTION 2:

A project manager is asked to deliver a project earlier than planned. Which actions are consistent with project crashing?

a. Increase the duration of critical activities.
b. Add more skilled personnel to critical tasks.
c. Use faster equipment or tools for critical activities.
d. Reduce the scope of the project.
e. Assign overtime to critical-path activities.

Correct Answer: b. Add more skilled personnel to critical tasks.
c. Use faster equipment or tools for critical activities.
e. Assign overtime to critical-path activities

Detailed Solution:

Project crashing is a schedule compression technique that aims to shorten the project duration without changing scope by adding resources or increasing efficiency, typically on critical-path activities. Adding skilled personnel (b), using faster tools/equipment (c), and assigning overtime (e) are classic crashing actions. Increasing activity duration (a) would lengthen the project, and reducing scope (d) is not part of standard crashing—it is scope reduction, not crashing.

QUESTION 3:

Which of the following conclusions are consistent with the Hawthorne Effect?

a. Human factors play a role in workplace productivity.
b. Worker productivity is independent of observation
c. Employee performance can be influenced by perceived interest from management.
d. Lighting conditions are the sole determinant of productivity.
e. Being part of a study may affect worker behavior.

Correct Answer: a. Human factors play a role in workplace productivity
c. Employee performance can be influenced by perceived interest from management
e. Being part of a study may affect worker behavior

Detailed Solution:

The Hawthorne Effect shows that workers' productivity is influenced by being observed and by perceived management interest, not just by physical conditions like lighting. This confirms that human factors (a), management attention (c), and awareness of being studied (e) all affect behavior. Observation does affect productivity (b is false), and lighting is not the sole determinant (d is false).

QUESTION 4:

Which of the following statements about a functional organization are true?

a. Functional organizations eliminate the need for management layers.
b. Functional groups often follow a pyramidal structure.
c. Functional organizations discourage technical specialization.
d. Functional organizations are commonly found in military, police, and hospitals.
e. Individuals are grouped according to specialized functions.

Correct Answer: b. Functional groups often follow a pyramidal structure.
d. Functional organizations are commonly found in military, police, and hospitals
e. Individuals are grouped according to specialized functions

Detailed Solution:

Functional organizations group individuals by specialized functions (e.g., engineering, marketing) and typically follow a pyramidal hierarchy with multiple management layers. They are common in military, police, and hospitals. They do not eliminate management layers (a), nor do they discourage technical specialization—they actually encourage it (c).

QUESTION 5:

According to Belbin, which of the following statements are correct?

a. An eligible candidate has the required qualifications.
b. A suitable candidate is capable of performing the job effectively.
c. Eligibility and suitability always imply the same thing.
d. A person may be eligible but not suitable.
e. A suitable candidate can never lack formal qualifications.

Correct Answer: a. An eligible candidate has the required qualifications.
b. A suitable candidate is capable of performing the job effectively
d. A person may be eligible but not suitable

Detailed Solution:

Belbin distinguishes eligibility (formal qualifications, experience) from suitability (behavioral attributes, team fit, and capability to perform effectively). A candidate can be eligible but not suitable (e.g., highly qualified but poor collaborator), and eligibility and suitability are not the same. Suitability does not require formal qualifications, so option (e) is false. Thus, a, b, and d are correct.

QUESTION 6:

Which of the following statements are consistent with Maslow's theory of motivation?

a. Food and shelter are examples of higher-level needs.
b. Human needs can be arranged in a hierarchy.
c. Higher-level needs emerge as lower-level needs are fulfilled.
d. All individuals are motivated by exactly the same needs.
e. Self-actualization is at the highest level of the hierarchy.

Correct Answer: b. Human needs can be arranged in a hierarchy.
c. Higher-level needs emerge as lower-level needs are fulfilled
e. Self-actualization is at the highest level of the hierarchy

Detailed Solution:

Maslow's theory proposes a hierarchy of human needs, where lower-level needs (physiological, safety) must be substantially satisfied before higher-level needs (esteem, self-actualization) emerge as motivators. Food and shelter are lower-level, not higher-level (a), and individuals differ in which needs they are pursuing (d). Self-actualization is indeed the highest level (e). Thus, b, c, and e are correct.

QUESTION 7:

Which statements correctly distinguish organization structure from team structure?(doubt on option a)

a. Organization structure focuses on teams within the organization.
b. Team structure focuses on individuals within a team.
c. Team structure determines how organizations are divided into departments.
d. Organization structure and team structure operate at different levels.
e. Team structure is unrelated to the arrangement of individuals.

Correct Answer: a. Organization structure focuses on teams within the organization.
b. Team structure focuses on individuals within a team.
d. Organization structure and team structure operate at different levels.

Detailed Solution:

Organization structure operates at the macro level, defining how the entire enterprise is divided into departments, reporting lines, and teams—making option (a) correct. Team structure works at the micro level, focusing on the roles and relationships of individuals within a specific team (b), and the two structures function at fundamentally different levels of analysis (d).

QUESTION 8:

According to Herzberg, motivators are factors that:

a. Make the job feel worthwhile.
b. Create a sense of achievement.
c. Eliminate all dissatisfaction.
d. Increase job satisfaction.
e. Include working conditions.

Correct Answer: a. Make the job feel worthwhile.
b. Create a sense of achievement.
d. Increase job satisfaction

Detailed Solution:

According to Herzberg's Two-Factor Theory, motivators (intrinsic factors like achievement, recognition, and meaningful work) make the job feel worthwhile (a), create a sense of achievement (b), and directly increase job satisfaction (d). They do not eliminate dissatisfaction—that is the role of hygiene factors (c), and working conditions are hygiene factors, not motivators (e).

QUESTION 9:

Which of the following statements are consistent with the Oldham–Hackman model?(doubt on option c)

a. A job requiring multiple skills may be perceived as more meaningful.
b. Seeing a whole piece of work contributes to task identity.
c. Understanding the importance of one's work relates to task significance.
d. Motivation depends solely on financial rewards.
e. Feedback can enhance job satisfaction.

Correct Answer: a. A job requiring multiple skills may be perceived as more meaningful.
b. Seeing a whole piece of work contributes to task identity.
e. Feedback can enhance job satisfaction.

Detailed Solution:

In the Oldham–Hackman model, skill variety (a) makes work feel more meaningful by engaging multiple talents, task identity (b) enhances meaning by allowing workers to see a complete, identifiable piece of work, and feedback (e) directly improves job satisfaction by providing clear information about performance. These are three of the five core job characteristics that foster intrinsic motivation, while the model does not rely on financial rewards alone, and task significance is specifically about the job's impact on others—not merely understanding importance.

QUESTION 10:

Which of the following are ways in which good project management can reduce stress?

a. Providing realistic effort estimates.
b. Reducing unexpected crises through better control.
c. Increasing uncertainty about responsibilities.
d. Reducing role ambiguity.
e. Encouraging conflicting responsibilities.

Correct Answer: a. Providing realistic effort estimates.
b. Reducing unexpected crises through better control
d. Reducing role ambiguity

Detailed Solution:

Good project management reduces stress by setting realistic effort estimates (a), minimizing unexpected crises through proactive control (b), and clarifying roles to reduce ambiguity (d). In contrast, increasing uncertainty about responsibilities (c) and encouraging conflicting responsibilities (e) would heighten stress, not reduce it.

******THE END******